Monologue or Dialogue?

Measuring Conversational Structure in the Reasoning Traces of Large Language Models

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Abstract

Reasoning models such as DeepSeek-R1 produce a long "thinking" trace before they answer. Such a trace can read as a single mind reasoning step by step, or as several voices trading proposals and objections. We ask which of these descriptions the text better fits when the question is put to measurement. There is no established ground truth for the voice structure of a reasoning trace, so we assemble a 2×2 grid of corpora whose structure *is* known (human and AI, monologue and dialogue) and let a detection method judge the reasoning traces only after it has told monologue from dialogue on all four known corpora. Three methods are tested, each reading a different textual signal: a document-level single- versus multi-author classifier (Nath, 2019), a sentence-level detector of where the writing style switches (Schmidt et al., 2025), and dialogue-act tagging that measures conversational structure through adjacency pairs (Duran et al., 2022; Schegloff and Sacks, 1973). All analysis decisions were fixed and placed under version control before any statistic was computed on the reasoning traces. [TODO: One sentence stating the verdict pattern: which methods passed the validity check, and whether R1 traces read as monologue, dialogue, or mixed.]

1 Introduction

Reasoning-oriented training has become standard in frontier language models: given a problem, a model such as DeepSeek-R1 (Guo et al., 2025) first generates an extended stretch of intermediate text, a *reasoning trace*, and only then commits to an answer. These traces have an unusual property. Each is produced by a single model in one generation pass, yet a trace often contains questions and the answers to them, objections and concessions, and abrupt reversals ("Wait, that's not right"). In text written by people, such features can mark a genuine exchange between participants, though they can equally mark a single vivid inner voice. This raises a question that is easy to state and less easy to settle: *does a reasoning trace have the textual structure of a monologue or of a dialogue?*

Kim et al. (2026) argue, drawing on both interpretability analysis and automatic labeling of the traces, that reasoning-trained models improve in part because they simulate a "society of thought": distinct internal perspectives that propose, criticize, and reconcile. Cognitive science offers older versions of the same idea: Mercier and Sperber (2017) hold that human reasoning is at root argumentative and social, and Minsky (1986) modeled the mind as many interacting agents. Against this background, the monologue-or-dialogue question becomes a test one can actually run: if reasoning traces carry the measurable structure of dialogue, that is evidence consistent with these accounts; if they do not, the impression of an inner conversation may owe more to the reader than to the text.

Our premise is that voice structure leaves marks in the writing itself. The society-of-thought account describes the internal perspectives as differing in role and character (Kim et al., 2026); if distinct perspectives take turns producing the text, they should also differ in how they write, in the small, largely involuntary habits (function-word use, sentence rhythm, dialogue-act moves) that authorship analysis uses to tell writers apart. We therefore treat the question operationally: a monologue should read as one author, a dialogue as several, and we ask which of these the reasoning traces resemble. This is not the setting these methods were built for: their "authors" are ordinarily different people, not perspectives within one model. Section 2 returns to what that borrowing does and does not license.

No corpus of reasoning traces comes labeled with its true voice structure, which is what makes the question hard to settle directly. We address this with the grid in Figure 1. We assemble four corpora whose structure is known: human monologue (think-aloud protocols), human dialogue (group deliberation), AI monologue (one model solving a puzzle alone), and AI dialogue (four instances of one model, each given a different role, debating), alongside the reasoning traces themselves. A method may report on the traces only after it has correctly separated monologue from dialogue on all four known corpora; a method that fails this check is reported as failing and set aside.

The three methods read different signals. Two come from authorship analysis and the PAN style-change task (Zangerle et al., 2025): one classifies a whole document as single- or multi-author from its function-word profile (Nath, 2019), while the other works sentence by sentence, estimating at each point whether the writing style has switched (Schmidt et al., 2025). The third comes from conversation analysis: it labels each utterance with its dialogue act and checks for adjacency pairs, a question followed by an answer, a proposal by its acceptance or refusal, which Schegloff and Sacks (1973) describe as a basic unit of conversational organization. The first two share this lineage but differ in mechanism; the third is unrelated to either. Because they read the text in different ways, agreement among them carries more weight than any one method alone.

Two further choices shape the design. First, we hold the task fixed: the AI monologue, AI dialogue, and reasoning-trace corpora all solve the same puzzle, Game of 24 (Yao et al., 2023). The methods examine style and structure rather than subject matter, so the risk of one keying on topic is small to begin with; holding the task constant removes it and keeps as many variables fixed as possible. Second, pre-registration: every analysis rule, threshold, and figure was fixed and version-controlled before any statistic was computed on the reasoning traces (Section 4.5).

Contributions. We (i) frame the monologue-or-dialogue question about reasoning traces as a measurable comparison against corpora of known voice structure; (ii) evaluate three established detection methods on matched human and AI, monologue and dialogue corpora, admitting each to the reasoning traces only after it passes a validity check on the known corpora; and (iii) [TODO: report the verdict on DeepSeek-R1 traces].

2 Related Work

Multi-author and style-change detection. Stylometry rests on the observation that unconscious habits, such as function-word frequencies and character n-grams, are stable within an author and vary between authors (Stamatatos, 2009). The PAN shared-task series has posed style-change detection since 2016: given a document, decide whether one or several authors wrote it, and where the switches occur (Zangerle et al., 2019, 2025). Our Method 1 is the threshold-based clustering approach of Nath (2019) from the PAN 2019 edition; our Method 2 is a sequential sentence-pair classifier submitted to the PAN 2025 edition (Schmidt

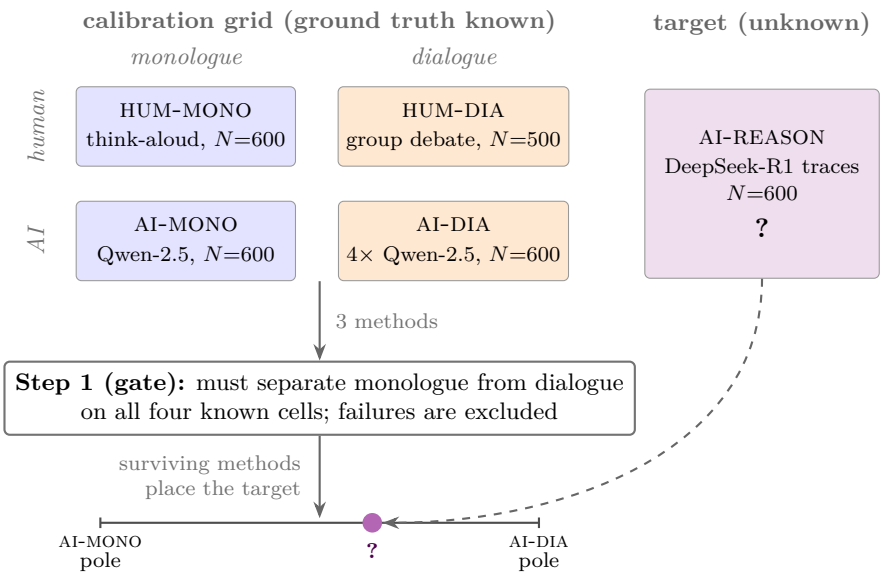


Figure 1: Study design. Four corpora with known voice structure (left) gate the three detection methods: a method may report on the target only after it separates monologue from dialogue on all four. Methods that pass then place the target on a monologue–dialogue axis anchored by the two same-task AI cells; the human cells are used only for validation, not for placement.

et al., 2025), built on content-independent style embeddings (Patel et al., 2025). Stylometric analysis of multi-authored documents more broadly is surveyed by Zamir et al. (2024). We repurpose this machinery for a question it was not designed for. In the original task the several authors are different people; here they would be perspectives within a single model's output. The justification is the premise stated in Section 1: the society-of-thought account attributes distinct roles and characters to these perspectives (Kim et al., 2026), and writers with distinct characters tend to leave distinct stylistic marks. Whether that assumption holds for one model's text is part of what the known-structure corpora are there to test.

Dialogue structure and conversation analysis. In conversation analysis, a dialogue is not just alternating text but a system of paired actions. Schegloff and Sacks (1973) describe the *adjacency pair*: two turns by different speakers in which the first makes the second relevant: a question makes an answer due, an offer makes an acceptance or a refusal due. This pairing is a basic way conversation is organized, and its two parts normally sit next to each other. Dialogue-act schemas make the pattern measurable by labeling each utterance with its function; we use the Conversation Analysis Modelling Schema (CAMS; Duran et al., 2022), which pairs a dialogue-act layer with adjacency-pair types. Adjacency-pair completion is the property that matters here, because a lone writer has little reason to produce it: a monologue may pose rhetorical questions, but it rarely answers them in the very next utterance the way a conversation does. Whether solitary human thought is itself monological is contested: in the Vygotskian tradition, inner speech is internalized dialogue and can keep the give-and-take of the conversations it grew from (Fernyhough, 2004; Alderson-Day and Fernyhough, 2015). Our design need not settle this: here "monologue" is defined by production, one speaker alone, and whatever dialogic residue a lone speaker's talk carries is what the known-structure corpora teach the methods to accept as a single voice.

Table 1: The five corpora ("cells"). The four calibration cells have known voice structure; the target does not. N = documents per cell.

Cell	Structure	Producer	Task	N
HUM-MONO	monologue	humans thinking aloud (Wurgaft et al., 2025)	Game of 24	600
HUM-DIA	dialogue	human groups, DeliData (Karadzhov et al., 2023)	Wason selection	500
AI-MONO	monologue	Qwen-2.5-7B, task-only prompt	Game of 24	600
AI-DIA	dialogue	4× Qwen-2.5-7B, stateless debate	Game of 24	600
AI-REASON	**unknown**	DeepSeek-R1, native thinking mode	Game of 24	600

Reasoning traces and their interpretation. Chain-of-thought prompting showed that intermediate text improves model problem-solving (Wei et al., 2022); reasoning-oriented training internalized the behavior (Guo et al., 2025). What the resulting traces *are* is contested. Kim et al. (2026) report that reasoning models engage diverse, sometimes conflicting internal perspectives, and that this shows up in the traces as conversational behavior: question-answer sequences, shifts in perspective, and the reconciliation of competing views. They reach this through several routes, including mechanistic interpretability and LLM-based labeling of the traces into distinct personas. Our study is narrower and complementary: we take a small set of established text-level methods, each with an explicit validity check on corpora of known structure, and ask only whether the traces sit closer to monologue or to dialogue. On the human side, think-aloud protocol analysis (Ericsson and Simon, 1984) treats verbalized solitary reasoning as data; Wurgaft et al. (2025) scale the method with automated transcription and coding, producing the Game-of-24 think-aloud corpus we use as our human-monologue cell. For human dialogue we draw on DeliData (Karadzhov et al., 2023), 500 group deliberations on the Wason card-selection task (Wason, 1968).

3 Study Design and Data

3.1 The calibration grid

The design is a 2×2 grid plus one target cell (Figure 1, Table 1). The grid cells vary producer (human vs. AI) and structure (monologue vs. dialogue), and for each the true structure is known, either by construction or from the source the corpus was collected from. The fifth cell, native DeepSeek-R1 reasoning traces, is the object of study.

The grid works as a gate: a method may report on the target only after it correctly separates monologue from dialogue on all four known cells (Section 4). A method that cannot tell the four voices apart in a genuine four-way debate is in no position to rule them out in a reasoning trace; a method that reports several voices in a lone person's think-aloud protocol is in no position to affirm them there.

3.2 Task control

A detection method can respond to what a text is about, not only to how many voices produced it, and the domain of a task can shape a text's structure. To keep such influences from being mistaken for voice structure, we hold the task fixed wherever possible: AI-MONO, AI-DIA, and AI-REASON all solve Game of 24 (Yao et al., 2023): combine four given numbers arithmetically to reach 24, and HUM-MONO uses the same puzzle, so a method reading human monologue as a single voice cannot be doing so merely in reaction to the puzzle. HUM-DIA is the exception: DeliData is built around the Wason card-selection task, and no Game-of-24 group-deliberation corpus exists. We chose the Wason task because it is likewise a compact

logic puzzle, the closest human group-deliberation data we could find to Game of 24. Its role is limited, to confirm that a method detects human dialogue at all, so we do not use it as a placement anchor (Section 4.4). The comparison that carries the placement is the same-task AI set: AI-MONO vs. AI-DIA vs. AI-REASON.

3.3 Generated cells

AI monologue (ai-mono). One call per puzzle to an instruction-tuned model (`qwen/qwen-2.5-7b-instruct`), with a minimal prompt that states the puzzle and its rules and asks for a solution, imposing no further structure on the response (Appendix B).

AI reasoning traces (ai-reason). One call per puzzle to DeepSeek-R1 (`deepseek/deepseek-r1`) in its native thinking mode, using the same minimal prompt as AI-MONO. The document is the model's reasoning trace, taken from the thinking channel of the response.

AI dialogue (ai-dia). This cell needed the most care, because the multiplicity has to come from separate model calls rather than from a single call asked to write several parts. Four instances of the same model (`qwen/qwen-2.5-7b-instruct`, as in AI-MONO) take turns writing the debate, one API call per turn. The instances are stateless and share only the running transcript: at each turn the current speaker sees the debate so far, the turns the other participants have written, and nothing else. Each is given a fixed role (creative problem solver, critical analyst, mathematical validator, alternative seeker) and a name, a letter A–D, so the participants can address one another. Turn order is drawn at random. A debate ends once all four participants have spoken and the exchange has either signalled a conclusion or begun to repeat itself, or at a hard cap of 40 turns; Appendix B gives the exact stopping rule. Two features of this setup are worth noting. We use four instances of one model rather than four different models, because the reasoning traces we study also come from a single model: if a method cannot find multiple voices in a same-model debate, that reflects the method's sensitivity, which the gate records. And no single call is ever asked to produce the whole four-person exchange, because a single call performing many voices is the very thing we are trying to detect in the reasoning traces and so cannot also serve as a positive control. The two anchors share one base model (Qwen-2.5-7B), while the target is a different model (DeepSeek-R1); Section 7 discusses the implications.

3.4 Acknowledged confounds

Several properties of the corpora could influence a method's output alongside voice structure; we state them here and return to them in Section 7. First, in the human pair, HUM-MONO and HUM-DIA differ in task as well as structure (Game of 24 vs. Wason), so a method's success in separating them could partly reflect task; this is why the human pair only validates and the AI set carries the placement. Second, the four debate roles in AI-DIA may add stylistic variation beyond multiplicity itself, though we take role differentiation to be part of what multiplicity means here, following the society-of-thought account (Kim et al., 2026), in which distinct perspectives are distinct roles; a positive style-method result on AI-DIA should be read as an upper bound on the role-free component. Third, HUM-MONO is transcribed speech while every AI cell is written text, so human–AI comparisons mix producer with register. Fourth, document lengths span two orders of magnitude across cells (Figure 2), and the reasoning traces are longer than either anchor; Section 4.2 shows where this matters and how we handle it.

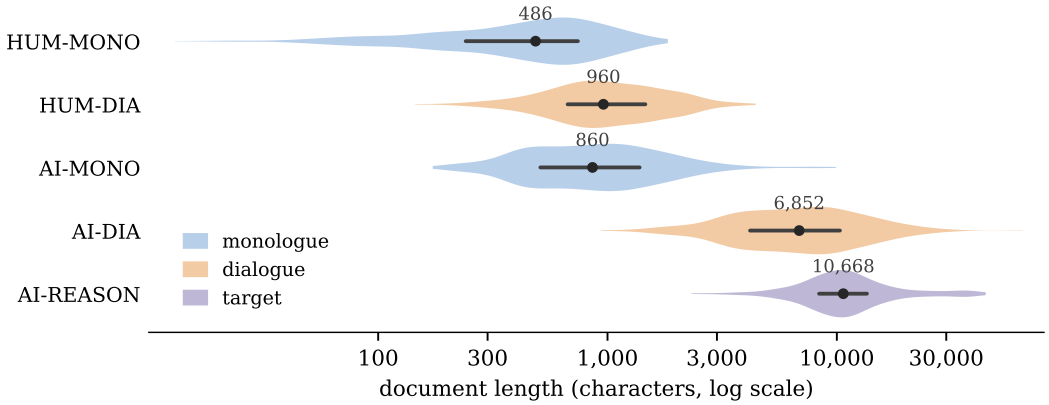


Figure 2: Document length per cell (log scale; violins with median dot, interquartile bar, and annotated median). Length differences follow from the design: dialogue produces more text than monologue, and the reasoning traces are longer than both AI anchors. Each cell shows its 600-document analysis sample (HUM-DIA: 500).

4 Methods

Each of the three methods follows the same procedure. First we calibrate it: run it on the four known cells and test it against a validity gate (defined per method below). A method that passes is then applied, unchanged, to the reasoning traces; a method that fails is reported and set aside. Reading a surviving method's result means placing the reasoning traces on a monologue–dialogue axis anchored by the two AI cells, AI-MONO and AI-DIA (Section 4.4). The three methods work on different signals: a function-word frequency clustering, a pretrained neural style encoder (with NLTK Punkt sentence splitting; Bird et al., 2009), and an LLM dialogue-act tagger, and are run independently of one another.

4.1 Method 1: Document-level style-change detection

Nath (2019) estimates how many authors wrote a document by clustering its paragraph-sized windows on their function-word profiles. The 50 most frequent tokens of the document define a small feature space, and each window is described by how often those tokens occur within it; windows are then compared by Matusita distance and grouped by threshold-based clustering (TBC), with the number of resulting clusters taken as the author count. We use the author's own implementation in the configuration it reports as best, and make two corrections so that our run matches the method as the paper describes it. The shipped script defaults to a different distance measure, which we set back to the Matusita distance the paper evaluates. And a filtering step in the code discards paragraphs shorter than 200 characters, whereas the paper specifies merging them into the preceding window; we restore the merging behavior, since our debate turns are often short and discarding them would remove exactly the turns that carry the dialogue structure.

Windowing follows the shape of each cell. In PAN's forum data a paragraph is one person's post, so for our turn-structured cells (AI-DIA, HUM-DIA) we treat one turn as one paragraph. The continuous-prose cells (HUM-MONO, AI-MONO, AI-REASON) have no such natural unit; for these we rely on the implementation's own handling, which splits a document on its blank lines and, where a document has none, as with transcribed speech in HUM-MONO, falls back to grouping ten sentences per window and splitting overlong paragraphs at five sentences.

Output. Each document receives a predicted author count, which we reduce to a binary call (multi-author if the count exceeds one). We evaluate accuracy pooled across the four

calibration cells and report the per-cell breakdown and a confusion matrix alongside. *Gate: the pooled binary accuracy must be significantly better than chance* (one-sided binomial test against 0.5, $p < .05$). We do not import the original paper's 0.85 accuracy as a threshold: that was its best score in its own domain (long StackExchange threads), and our data are short puzzle transcripts, so we ask only that the method distinguish single- from multi-author text better than chance.

4.2 Method 2: Sentence-level style-change detection

The sequential sentence-pair classifier of Schmidt et al. (2025) encodes each sentence with a content-independent style embedding (Patel et al., 2025), contextualizes the sequence with a BiLSTM, and emits, for every adjacent sentence pair, the probability that the two sentences were written by different authors. We run the shipped pretrained checkpoint as is, which amounts to out-of-domain inference, with no fine-tuning on our data, and read out the raw per-pair probabilities before any thresholding. The per-document raw score is the mean of these probabilities: a scalar in $[0, 1]$, higher meaning more style switching.

Length. The reasoning traces are much longer than either anchor (median 10,668 characters vs. 6,852 for AI-DIA and 860 for AI-MONO; Figure 2), and length affects the raw mean directly. A longer document is divided into more sentence pairs, so if a document carries some roughly fixed number of genuine style switches, spreading them over more pairs lowers the mean; a raw sum has the opposite problem, growing with length even at a constant rate of chance switches. Rather than choose between the two, we model the length relationship on the calibration data. For each anchor separately, using only that anchor's documents, we fit a curve of switch probability against document length, choosing among linear, semi-log, and log-log forms by AIC (Akaike, 1974), so that the two anchors need not share a functional form. Each document then receives two residual distances: d_{dia}, its deviation from the AI-DIA curve at its own length, and d_{mono}, its deviation from the AI-MONO curve. The per-document score, which we call the *lean*, is $\ell = |d_{\text{mono}}| - |d_{\text{dia}}|$; a positive value means the document sits closer to the dialogue curve. Because the reasoning traces run past both anchors' observed length ranges, we show the extrapolation directly (Figure 5).

Gate. Mean lean for the dialogue cells (HUM-DIA, AI-DIA) must exceed that for the monologue cells (HUM-MONO, AI-MONO), with Cliff's $\delta \geq 0.33$ (Cliff, 1993) between the two groups, a direction check with a medium-effect-size floor, so that a technically correct but negligible separation does not count as validation.

4.3 Method 3: Dialogue-act structure

Every document is segmented into utterances, and each utterance is labeled with one of the 27 CAMS dialogue-act tags (Duran et al., 2022), plus `none`, by an LLM tagger with a fixed prompt at temperature 0 (Appendix A). For example, "Wait, does 6×4 actually give 24?" is tagged `checkQuestion`, and a following "Yes, that works" is tagged `confirm`. We use only the dialogue-act layer of CAMS: its formal adjacency-pair annotation layer requires speaker-turn structure that three of our five cells do not have. From the tag sequences we compute three signals of increasing stringency, identically for every cell.

(a) Tag distribution. For each cell, the normalized frequency of each of the 28 labels across all of its utterances, giving a 28-dimensional profile. Does the reasoning traces' mix of conversational functions resemble the monologue cells or the dialogue cells?

(b) Tag transitions. The 28×28 matrix of counts of one tag immediately following another. Because a transition must stay within a single document (the last tag of one document does not precede the first tag of the next), these counts are formed per document and then summed within each cell: which conversational move tends to follow which?

Table 2: Mapping from first pair-part (FPP) tags to valid second pair-part (SPP) tags, derived from CAMS's category structure (Duran et al., 2022) and fixed before any data was seen.

FPP tag	Valid SPP tag(s)	CAMS basis
`propositionalQuestion`	`answer`, `confirm`, `disconfirm`	yes/no question → answer
`setQuestion`	`answer`	wh-question → answer
`choiceQuestion`	`answer`	alternative question → answer
`checkQuestion`	`confirm`, `disconfirm`	belief check → (dis)confirmation
`request`	`accept`, `decline`, `conditionalAccept`	directive → commissive
`suggest`	`accept`, `decline`, `conditionalAccept`	directive → commissive
`offer`	`accept`, `decline`, `conditionalAccept`	offer → accept/decline
`inform`	`agreement`, `disagreement`, `correction`, `autoPositive`, `autoNegative`	statement → feedback
`greeting`	`greeting`	social-obligation pair
`goodbye`	`goodbye`	social-obligation pair
`thanking`	`acceptThanking`	social-obligation pair
`apology`	`acceptApology`	social-obligation pair

(c) Adjacency-pair completion (primary signal). Signals (a) and (b) are profiles; a monologue that merely *contains* many questions could score dialogue-like on them. Signal (c) tests what adjacency-pair theory identifies as the mark of conversation: when a first pair-part (FPP) occurs (a question, request, suggestion, offer), is it met by a valid second pair-part (SPP) in the *very next* utterance? The mapping from each FPP tag to its valid responses (Table 2) is derived from CAMS's category structure and was fixed before any data was seen. The per-document metric is the *completion rate*: of all adjacent tag pairs whose first member is an FPP, the fraction whose second member is a valid SPP. A document that contains no FPP has a completion rate of 0 (it initiates no pairs to complete); we also report, per cell, the share of such documents, since a monologue cell is expected to contain many. We require immediate adjacency: the SPP in the very next utterance, with no inserted material, which CAMS allows but which needs speaker structure to detect. This keeps the measure conservative: by ignoring completions separated by an insertion, it can only understate how often pairs are completed, so any dialogue-ward difference it produces is not an artifact of a lenient criterion.[TODO: FPP-absent documents are scored 0 here; the frozen `methods.md` instead excludes them and reports the absence share separately. Log this as a deviation, or reconcile the two.]

Gates. Method 3 must clear two gates. *Reliability*: a random sample of the tagger's labels from each calibration cell is checked against hand labels; Cohen's $\kappa \geq 0.70$ (Cohen, 1960) is required. We compute the sample size per cell from the asymptotic variance of κ (Fleiss et al., 1969): using each cell's own observed tag marginals, we solve for the n at which the 95% confidence interval around the measured κ has half-width ≤ 0.05, so that a pass/fail call near the cutoff reflects the tagger rather than a too-small sample. *Validity*: the completion rate (signal c) must separate the dialogue cells from the monologue cells in the expected direction, with Cliff's δ reported. If either gate fails, Method 3 is reported as failed and makes no call on the target.

4.4 Placing the target and combining methods

Each surviving method places AI-REASON by the same nearest-pole rule, with AI-MONO as the monologue anchor and AI-DIA as the dialogue anchor (the human cells validate; they do not place, Section 3.2). For scalar signals the distance to a pole is the absolute difference of cell means; for the distributional signals of Method 3 it is the Jensen–Shannon divergence (Lin,

Table 3: Method 1 calibration and target summary. Gate: pooled binary accuracy vs. chance, one-sided binomial test.

Cell	Mean author count	Share called multi-author	Accuracy
HUM-MONO	⟨x⟩	⟨x⟩	⟨x⟩
HUM-DIA	⟨x⟩	⟨x⟩	⟨x⟩
AI-MONO	⟨x⟩	⟨x⟩	⟨x⟩
AI-DIA	⟨x⟩	⟨x⟩	⟨x⟩
Pooled (gate)	—	—	⟨x⟩, $p = ⟨x⟩$
AI-REASON (target)	⟨x⟩	⟨x⟩	—

1991), symmetric, bounded, and defined under zeros. The verdict is *monologue-leaning* or *dialogue-leaning* toward the nearer pole, unless the two distances fall within a relative margin m of one another, in which case the verdict is *no clear margin*. The margin is calibrated as follows. For each signal, we take repeated random same-pole splits of AI-DIA and AI-MONO (1,000 bootstrap iterations each), giving the distribution of relative gaps that sampling noise alone produces between two halves of one cell; m is the 95th percentile of that distribution. A gap between the reasoning traces and a pole counts only if it exceeds what 95% of these same-pole splits produce. Methods 1 and 2 each contribute one verdict; Method 3's verdict comes from its primary signal (c), with the two distributional signals reported alongside.

We report the pattern of the surviving methods' verdicts in full. If the methods agree, the shared verdict is the finding, and its weight comes from three differently constructed measures pointing the same way. If they disagree, we report the disagreement and read it as mixed evidence, some structure of each kind in the reasoning traces, rather than resolving it by counting verdicts.

4.5 Pre-registration

All rules above (method configurations, gates, the FPP–SPP table, the placement rule, and the exact set of figures and tables in Section 5) were fixed in a written specification and committed to version control before any summary statistic was computed on the reasoning traces. The methods are taken from their source papers and were not tuned on our data; the known-structure cells serve to check each method's validity, not to fit it. One point deserves precision: producing the reasoning traces' raw per-utterance tags (Method 3) is a temperature-0 labeling pass with no tunable choice, and was done before the specification was frozen; every summary statistic over those tags (distributions, transitions, completion rates, divergences) was computed only afterward. Any later deviation is recorded with its reason rather than edited into the specification. The pre-registration commit is `f2c300d9`, dated 2026-07-16.

5 Results

[TODO: This section is a pre-registered skeleton: every figure and table below was specified before the numbers existed. Fill in as analyses complete; nothing may be added or dropped without a logged deviation.]

5.1 Method 1: Style-change detection

[TODO: Lead-in: pooled accuracy ⟨acc⟩, binomial $p = ⟨p⟩$; gate passed/failed.]

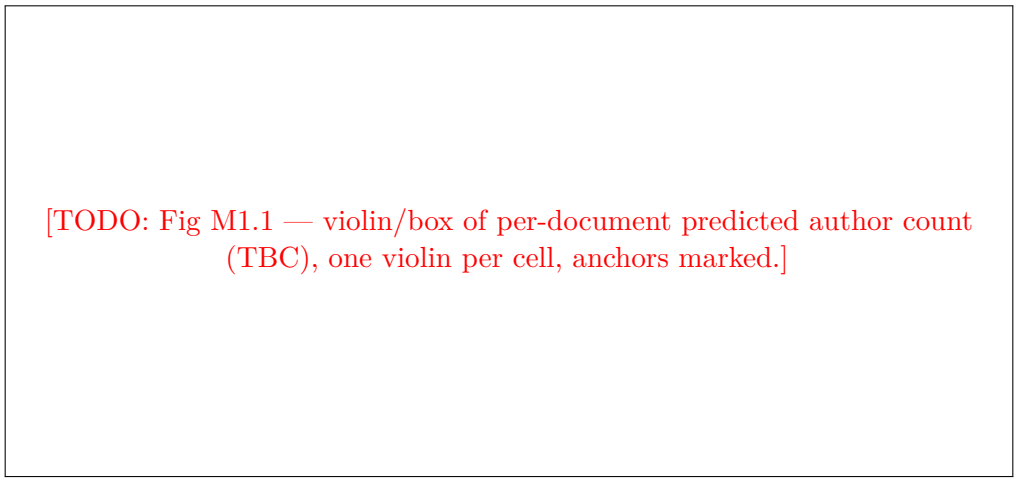


Figure 3: Predicted author count per document, by cell (Method 1, TBC). Reference lines mark the anchors AI-MONO and AI-DIA; the target is highlighted. [TODO: finalize caption]

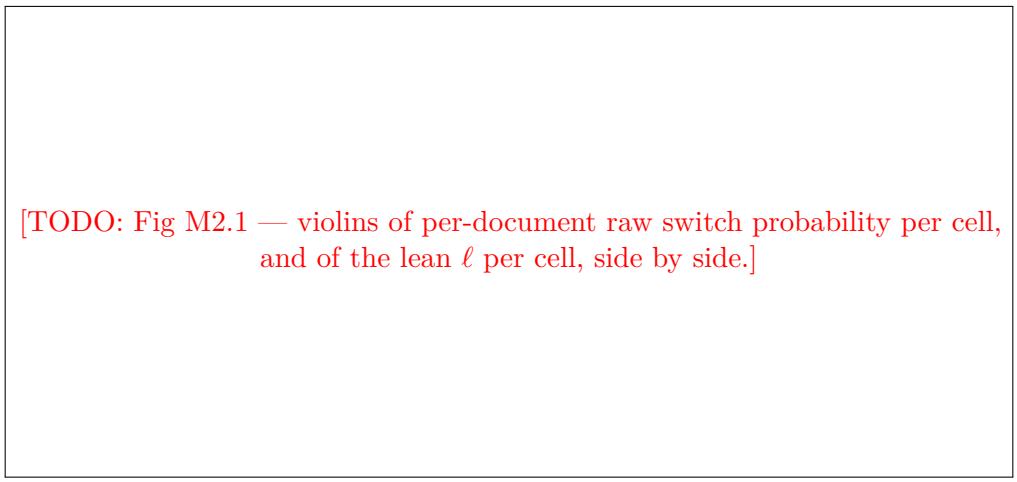


Figure 4: Left: raw mean switch probability per document, by cell. Right: the length-aware lean $\ell = |d_{\text{mono}}| - |d_{\text{dia}}|$ (positive: closer to the dialogue curve). [TODO: finalize caption]

5.2 Method 2: Sentence-level switch probability

[TODO: Lead-in: chosen functional form per anchor (AIC), gate direction and Cliff's δ = ⟨delta⟩; gate passed/failed.]

5.3 Method 3: Dialogue-act structure

[TODO: Lead-in: κ = ⟨kappa⟩ (CI), reliability gate; validity gate direction and Cliff's δ; then the three signals.]

5.4 Cross-method summary

6 Discussion

[TODO: Write after results. The frame below was fixed in advance so the interpretation cannot drift toward whatever the numbers happen to say.]

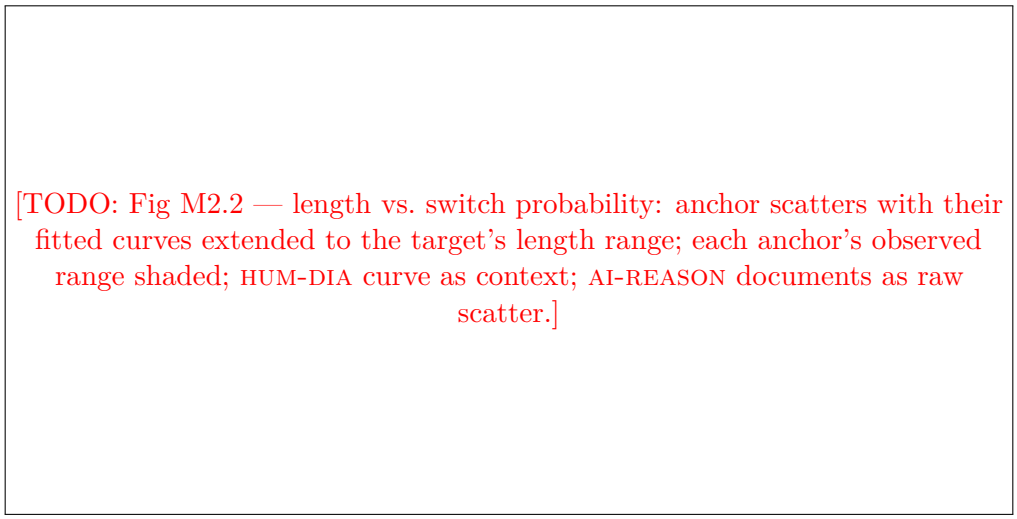


Figure 5: The length–probability relationship and the extrapolation it requires. Each anchor's curve is fit on its own documents only (functional form chosen by AIC per anchor); shaded bands mark each anchor's observed length range, so the stretch to the target's lengths is visible rather than asserted. The HUM-DIA curve is context only; the target's documents appear as a raw scatter, not a fit. [TODO: finalize caption; annotate how far past each anchor's maximum length the target extends]

Table 4: Method 2 summary: per-cell mean lean, gate effect size, and the per-anchor functional-form comparison.

	Value
Mean lean ℓ: HUM-MONO	⟨x⟩
HUM-DIA	⟨x⟩
AI-MONO	⟨x⟩
AI-DIA	⟨x⟩
AI-REASON	⟨x⟩
Gate: direction (HUM-DIAAI-DIA > HUM-MONOAI-MONO)	⟨pass?⟩
Gate: Cliff's δ (floor 0.33)	⟨x⟩
AIC, AI-DIA curve (linear / semi-log / log-log)	⟨x⟩ / ⟨x⟩ / ⟨x⟩
AIC, AI-MONO curve (linear / semi-log / log-log)	⟨x⟩ / ⟨x⟩ / ⟨x⟩

What a verdict does and does not mean. Whatever the outcome, the claim this study licenses is about *textual surface structure*: whether reasoning traces exhibit the style-switching and adjacency-pair regularities characteristic of multi-party dialogue. A dialogue-leaning verdict would not show that a model contains multiple minds, any more than a novelist's dialogue does; it would show that the trace is organized as conversation, itself a substantive, measurable fact, and the surface counterpart of the internal multi-agent dynamics reported by Kim et al. (2026). A monologue-leaning verdict would suggest the conversational flavor of reasoning traces is thinner than it reads: local markers ("wait," "but") without the structural commitments of an actual exchange. A split between the methods would itself be diagnostic. Dialogue-like functional structure without stylistic voice changes would describe a single narrator performing an argumentative script; stylistic switching without completed adjacency pairs would describe register drift rather than conversation. Each outcome is informative; none was privileged in advance.



Figure 6: Dialogue-act tag distributions (signal a), by cell. [TODO: finalize caption]

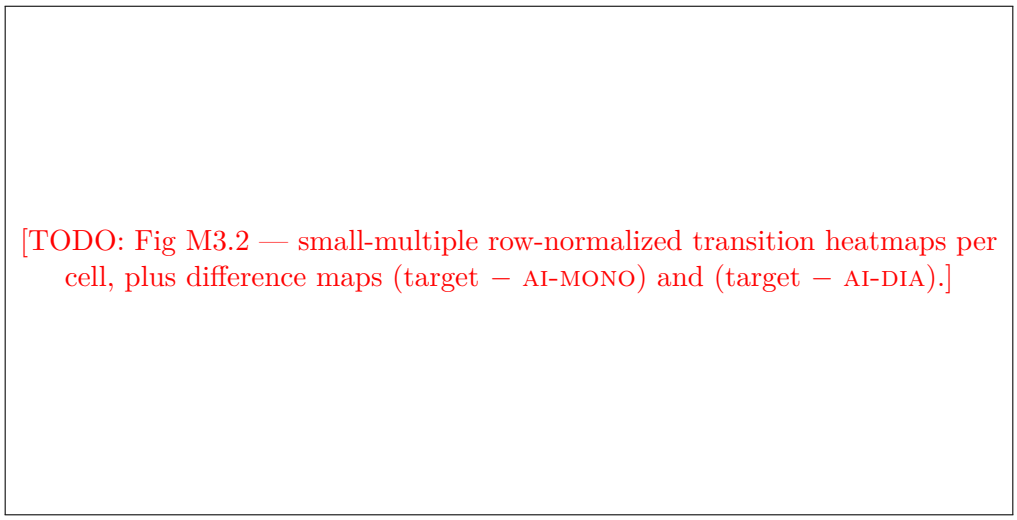


Figure 7: Tag transition structure (signal b): per-cell transition matrices and two difference maps showing which transitions pull the target toward each anchor. [TODO: finalize caption]

Interpretation of the observed pattern. [TODO: Fill in: which methods passed, what pattern emerged, what it means under the frame above. Connect to Kim et al.'s internal evidence: does the outside view agree with the inside view, or do internal dynamics and surface text dissociate?]

7 Limitations

One model, one task. The target is one reasoning model (DeepSeek-R1) on one puzzle family (Game of 24). Task control bought interpretability at the price of generality; the design transfers directly to other models and tasks, but the present verdict is about this cell, not about reasoning traces at large.

The target is a different base model than the anchors. Both anchors were generated with Qwen-2.5-7B, which keeps the monologue–dialogue contrast between them clean: the only intended difference is the number of participants. The target, however, is DeepSeek-R1, so part of any measured distance between target and anchor may reflect base-model register rather than voice structure. This is unavoidable: the object of study is R1's native traces,

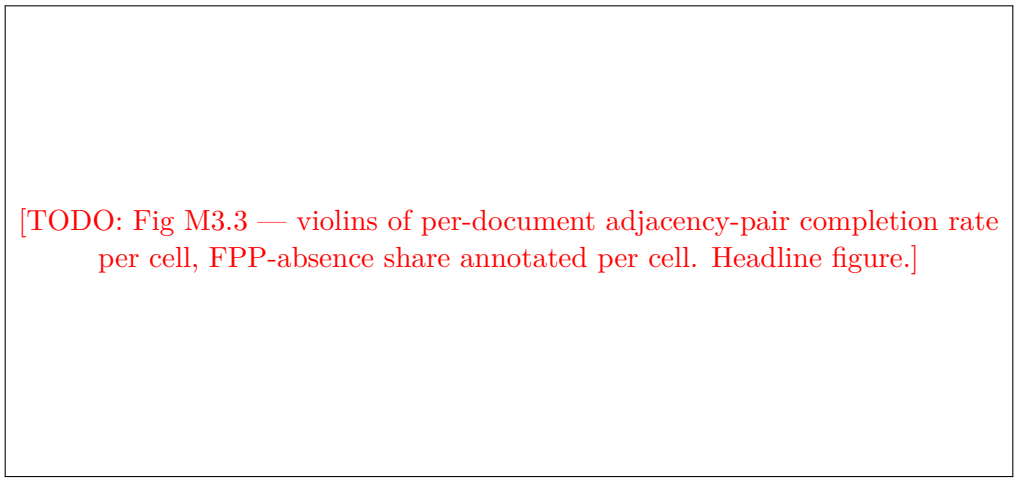


Figure 8: Adjacency-pair completion rate (signal c, primary), by cell. Annotations give each cell's share of documents containing no FPP at all. [TODO: finalize caption]

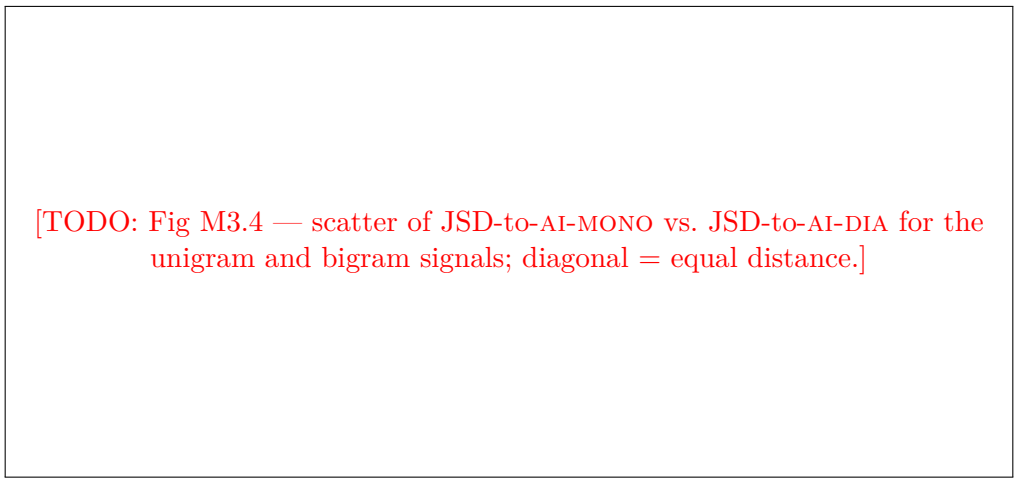


Figure 9: Distributional placement (signals a, b): each cell's divergence from the monologue anchor (x) against its divergence from the dialogue anchor (y). Points below the diagonal lean dialogue. [TODO: finalize caption]

and no "non-reasoning R1" exists to serve as its own anchor. It means the placement should be read as *closer to which pole*, where both poles carry the same model offset, not as an absolute distance.

Role-induced style in the AI debate. The four debate participants were given distinct roles. Style-based methods may therefore separate AI-DIA from AI-MONO partly on role-driven register rather than multiplicity alone; AI-DIA validation results are an upper bound in that respect. The dialogue-act method is less exposed to this, since it keys on the sequence of dialogue-act functions rather than on stylistic contrast between speakers.

Stability of the debate identities. Because turn order in AI-DIA is random, a participant is sometimes selected to speak twice in a row, and a model instance occasionally treats its own earlier turn as another participant's. The role assignments are therefore not perfectly stable across a debate. This does not affect the cell's use here, since it still contains genuine multi-call, multi-role text, but it is a caveat for anyone reusing the debate corpus on its own.

Length and extrapolation. The reasoning traces are longer than both anchors (Fig-

Table 5: Method 3 summary: reliability, per-cell completion rates, and the placement distances and calls per signal.

	Completion rate		Placement (target)	
Cell / signal	mean	FPP-absent share	d_{mono} / d_{dia}	call
HUM-MONO	⟨x⟩	⟨x⟩	—	—
HUM-DIA	⟨x⟩	⟨x⟩	—	—
AI-MONO	⟨x⟩	⟨x⟩	—	—
AI-DIA	⟨x⟩	⟨x⟩	—	—
AI-REASON	⟨x⟩	⟨x⟩	—	—
Signal a (unigram)	—	—	⟨x⟩ / ⟨x⟩	⟨call⟩
Signal b (bigram)	—	—	⟨x⟩ / ⟨x⟩	⟨call⟩
Signal c (completion)	—	—	⟨x⟩ / ⟨x⟩	⟨call⟩
Reliability	κ = ⟨x⟩ (95% CI ⟨x⟩), gate ≥ 0.70: ⟨pass?⟩			

[TODO: Fig X.1 — single placement panel: monologue–dialogue axis normalized so AI-MONO = 0 and AI-DIA = 1; each surviving method's target position as a point with its no-clear-margin band shaded.]

Figure 10: The answer in one view: each surviving method's placement of the target on the monologue–dialogue axis, with its calibrated no-clear-margin band. [TODO: finalize caption]

ure 2), and Method 2's placement requires evaluating anchor curves beyond their fitted ranges. We show the extrapolation explicitly (Fig. 5), but it remains extrapolation.

The human cells differ from the AI cells in more than voice structure. HUM-DIA deliberates the Wason task, not Game of 24, and HUM-MONO is transcribed speech while every AI cell is written text. Both cells therefore serve only to validate that a method detects human monologue and dialogue at all; neither anchors the placement.

Out-of-domain inference throughout. Methods 1 and 2 were built for forum text and essays, not puzzle transcripts; Method 3's tagger is an LLM applied outside CAMS's original annotation setting. The validity check exists for this reason: a method that survives it has shown competence on this domain.

LLM-based tagging. Method 3's labels come from an LLM. The reliability gate (κ against hand labels, at a computed sample size) bounds this risk but cannot rule out correlated errors that an LLM tagger might repeat consistently across cells.

8 Conclusion

[TODO: Three or four sentences: the question, the design in one line, the verdict pattern, and the one-sentence takeaway about how reasoning traces should be read. Write last.]

Table 6: Cross-method verdicts. This table answers the research question.

Method	Passed calibration?	Verdict on AI-REASON	Margin
1: Style change (document)	⟨x⟩	⟨x⟩	⟨x⟩
2: Style change (sentence)	⟨x⟩	⟨x⟩	⟨x⟩
3: Dialogue acts	⟨x⟩	⟨x⟩	⟨x⟩

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A CAMS tag set and tagging protocol

Tag set. The 27 CAMS dialogue-act tags (Duran et al., 2022), derived from the Dialogue Act Mark-up Language (DiAML, ISO 24617-2), grouped by function:

- *Information-seeking:* `propositionalQuestion` (yes/no), `setQuestion` (wh-), `choiceQuestion` (alternatives), `checkQuestion` (verifying an uncertain belief)
- *Information-providing:* `inform` (statement), `answer`, `agreement`, `disagreement`, `correction`, `confirm`, `disconfirm`
- *Commissive:* `offer`, `conditionalAccept`, `accept`, `decline`
- *Directive:* `request`, `suggest`
- *Feedback:* `autoPositive` (positive understanding), `autoNegative` (negative understanding)
- *Time management:* `stalling` (pausing)
- *Own-communication management:* `retraction` (abandoning an utterance)
- *Social obligation:* `greeting`, `goodbye`, `thanking`, `acceptThanking`, `apology`, `acceptApology`

Utterances fitting no tag receive `none`, giving 28 labels in total.

Tagging protocol. Each document is sent to the tagger LLM (`tencent/hy3`, via OpenRouter) in a single call at temperature 0, together with a fixed instruction prompt that reproduces CAMS's own labeling instructions: tag definitions with worked examples, annotation guidelines, and the full label inventory. The prompt file, `CAMS_LLM_labeling_prompt.txt`, is part of the project repository. The same call performs utterance segmentation and tagging: the model returns the document split into utterances, each with exactly one label. No per-cell adaptation of the prompt exists; the identical prompt and parameters are used for all five cells. Only the dialogue-act layer of CAMS is used; its adjacency-pair layer (FPP/SPP with expansions) requires speaker-turn structure that three of the five cells lack (Section 4.3).

B Generation details for the AI cells

Puzzle selection. Game-of-24 items were drawn from the problem set of the think-aloud corpus (Wurgaft et al., 2025), which supplies per-puzzle metadata including median human solving time and failure rate. From the solvable puzzles a subset of 200 was selected at random (fixed seed) and carried, with this metadata, into `puzzle_selection_game24.csv`; each generated cell covers these puzzles in three passes with distinct seeds, yielding 600 documents per cell. [TODO: Confirm the selection was random rather than difficulty-ranked: the generation script samples the solvable puzzles at random, but verify this is the run that produced the final data.]

AI monologue (ai-mono). Model `qwen/qwen-2.5-7b-instruct` via OpenRouter, one call per document. The complete user prompt, verbatim:

> `Target 24 with numbers [`n_1`, `n_2`, `n_3`, `n_4`]. Use each number exactly once with +, -, *, /.`

AI dialogue (ai-dia). Model `qwen/qwen-2.5-7b-instruct` via OpenRouter, the same model as AI-MONO. Four stateless participants, A–D, each realized as a separate API call per turn; a call receives the participant's own system prompt and the shared public transcript so far. The system prompt assigns the four fixed roles:

```
You are {X}, one of four participants in a mathematical problem-solving debate:
- A: Creative Problem Solver - proposes novel approaches, thinks outside the box
- B: Critical Analyst - challenges assumptions, questions logic, looks for flaws
- C: Mathematical Validator - verifies calculations step-by-step, checks correctness
- D: Alternative Seeker - looks for different solution paths, explores variations
Share your thought briefly while staying in character.  This is a collaborative but
critical discussion.
```

Turn order is drawn at random (seed recorded per debate). Turn boundaries are set by stop sequences on the speaker labels (`\nA:` … `\nD:`); temperature and maximum tokens are left at the provider defaults. A debate ends once all four participants have spoken and at least four turns have been produced, and then either (i) one of the last two turns contains a conclusion cue (*done*, *finished*, *complete*, *agreed*, *consensus*, *locked in*, *final*), or (ii) the recent turns have begun to repeat, detected as a TF–IDF cosine similarity above 0.92 between consecutive turns in the last few. Failing these, the debate stops at a hard cap of 40 turns. No separate final-answer round is collected; if an answer appears, it appears within the discussion.

AI reasoning traces (ai-reason). Model `deepseek/deepseek-r1` via OpenRouter, one call per puzzle with the same minimal prompt as AI-MONO; the document is the content of the model's native thinking channel, not its final answer.

Regeneration of corrupted debates. In the original AI-DIA generation run, 46 of 600 debates drifted into Chinese mid-conversation. These debates were regenerated in full, and the corpus was verified English-only afterward; both the original and regenerated files are retained, and the processed cell derives from the regenerated version. [TODO: Data check before Step 1: the API-reported `model_version` field marks 47 of the 600 processed debates as `deepseek/deepseek-chat-v3` rather than Qwen; apparently the regeneration pass ran with the script's default model. Either regenerate these 47 debates with `qwen/qwen-2.5-7b-instruct` (permissible: a calibration cell, before any Step-1 statistic) and delete this note, or disclose the mixture here and in Section 3.3.] One puzzle item was removed from AI-REASON because its answer could not be reliably extracted.

C Preprocessing summary

All five cells pass through a shared standardization pipeline (`standardize_datasets.py`) that produces the analysis-ready `processed_*.jsonl` files; originals are preserved unchanged, and every row carries a `processing_info` record documenting exactly what was altered (characters removed, LaTeX conversions, speaker labels stripped, timestamps). The pipeline's goal is a uniform character set and format, so that no method can separate cells on collection-pipeline artifacts rather than voice structure.

Shared normalization. Processing order: (1) extract structural metadata (system lines, speaker information) to dedicated fields; (2) remove speaker labels from transcript text; (3) convert LaTeX to plain-text math (`\frac{a}{b}` → `(a/b)`, `\times` → ×, delimiters and `boxed` artifacts removed); (4) filter to a fixed allowed character set (letters, digits, basic punctuation, arithmetic symbols); (5) normalize quotes (curly → ASCII) and whitespace. Characters outside the allowed set, such as foreign scripts, emoji, and decorative Unicode,

are removed; curly quotes are mapped to their ASCII equivalents so that the character set is uniform. Two choices are worth noting. Document length and turn count are not equalized across cells, since longer dialogues follow from the monologue/dialogue manipulation itself (length is instead handled at analysis time, Section 4.2); and the already-matched AI cells receive only the normalization needed to address a concrete artifact, to avoid removing real signal.

Per-cell notes. *HUM-MONO* (600 documents): a 600-document analysis sample drawn from the think-aloud corpus collected from 640 participants by Wurgaft et al. (2025); puzzle identifiers standardized to a common format; fields not present in the original human data (seeds, API costs) were removed. [TODO: The processed human-monologue file currently holds the full 5,143 rows; fix the exact 600-row analysis subset and confirm the figures use it.] *HUM-DIA* (500 documents): DeliData group deliberations (Karadzhov et al., 2023) converted from per-group TSV to JSONL; speaker labels (participant pseudonyms) stripped from the transcript text and recorded as removed in `processing_info`, so no speaker identity is recoverable downstream, a property reported in Section 4. *AI-DIA* (600 documents): the regenerated, verified-English debate corpus (Appendix B); turns retain per-turn speaker letters in structured fields, while the flat transcript field is label-free. *AI-MONO* (600 documents) and *AI-REASON* (600 documents): single-blob transcripts; LaTeX normalization is the main transformation, as both models emit mathematical markup.